

Array

Subject: C Language

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Introduction to Arrays :-

- Array is a linear Collection of similar elements.
- Array is also known as subscript variable.

Array Declaration Rules :-

- i) Array ko declare karte samaye empty nahi chor sakte otherwise wah error dega.

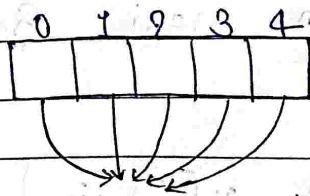
Ex:- `int a[ ];` // Error

- ii) `int a[5];` → i) yaha pe sirf natural no. likhte hain.
ii) Yaha per yeh number yah batata hai ki total number of variables kitne hai array ke andar.
iii) yeh index nahi hota hai.

NOTE:- Whatever is the size of an array it always consumes memory in a Sequential fashion.

- iii) Agar hum array ko initialize nahi karte sirf declare karke chor dete hai to automatic array ke har index pe garbage value assign ho jata hai.

Ex:- `int a[5];`



iv) Aap chahate to initialize kar sakte hai array ko Declare karte samaye.

Ex:- `int a[5] = {10, 50, 30, 70, 90};`

10	50	30	70	90
0	1	2	3	4

v) Aap array ko uske size se jayada nahi initialize kar sakte.

Ex:- `int a[5] = {10, 50, 30, 70, 90, 80, 40};` // Error

vi) Aap array ko uske size se kam value initialize kar sakte hai Declare karte samaye hi.

Ex:- `int a[5] = {10, 50};`

10	50	0	0	0
0	1	2	3	4

But: array ke Remaining variable ke andar zero (0) value assign ho jayega. Garbage value nahi.

vii) Array ko declare karte samaye aap empty chor sakte hai sirf ek hi Condition pe jab aap ussi samaye array ko initialize kar de.

Ex:- `int a[] = {10, 50, 30, 80, 20};`

Bound Checking

- i) C language me sirf Declaration Statement ko bound checking kiya jata hai But Action Statement ko nahi. So always be Carefull when writting a Code:

Ex:- a.) `int a[5] = { 10, 20, 50, 90, 80, 60, 70 };`

⇒ Iss example me basically array ke size se jayada element initialize kiya ja raha hai So yaha pe C language bound checking ke through error de dega beccause ye declaration statement hai.

b.) `int i, a[5];
for (i=0; i<=4; i++)
{
 scanf ("%d", &a[i]);
}`

⇒ But in this example yah action statement hai, So yaha pe C language bound checking nahi kar sakta isiliye Nah yaha pe error nahi dega but iss Code me error hai. So isiliye Code likhte Samay be Carefull rabe.

Sorting

- Arranging elements in some logical order is known as Sorting.
- By default, for numbers sorting means arranging elements in ascending order.
0 1 2 3 4 5 6 7
Ex:- Given Array $\rightarrow$ 20 50 40 60 70 80 30 10
Sorted Array $\rightarrow$ 10 20 30 40 50 60 70 80
- By default, for string sorting means arranging elements in alphabetical order.

Function Call by Passing array

```
int main()
```

```
{  
    int a[10];
```

```
    input(a, 10);
```

```
}
```

```
void input(int b[], int n)
```

```
{
```

```
    int i;
```

```
    printf("Enter %d numbers", n);
```

```
    for(i=0; i<n; i++)
```

```
    {
```

```
        scanf("%d", &b[i]);
```

```
    }
```

```
}
```


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Two Dimensional Arrays

⇒ A two-dimensional array is an array of arrays.

It stores data in the form of rows and columns, just like a table or matrix.

Declaration of 2D Array

```
int arr[3][4];
```

→ It means `int arr[row][column];`

Initialization of 2D Array

```
int arr[3][4] = {  
    0 {10, 20, 30, 40},  
    1 {50, 60, 70, 80},  
    2 {90, 100, 110, 120}  
};
```

Accessing Elements (using Loop)

```
for (int i = 0; i < 3; i++) {  
    for (int j = 0; j < 4; j++) {  
        printf("%d ", arr[i][j]);  
    }  
    printf("\n");  
}
```